Homework 4: Submission 2

Mathew Kang

Question 1

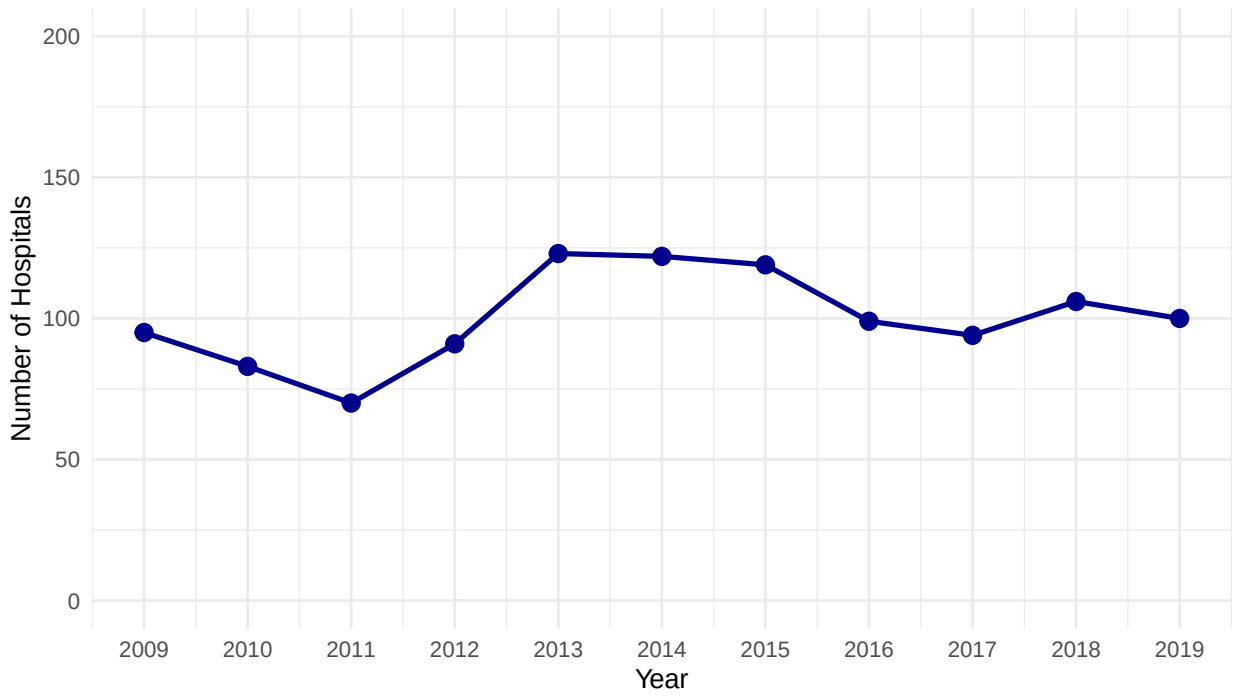


Figure 1: Hospitals That Filed More Than One Report In The Same Year (2009-2019)

For this question, because the final HCRIS dataframe contains only unique reports, I used the raw HCRIS data files for versions 1996 and 2010. Figure 1 presents the number of hospitals that filed more than one report in the same year, which shows a noticeable decrease in reporting from 2009 to 2011 before increasing again in 2012.

Question 2

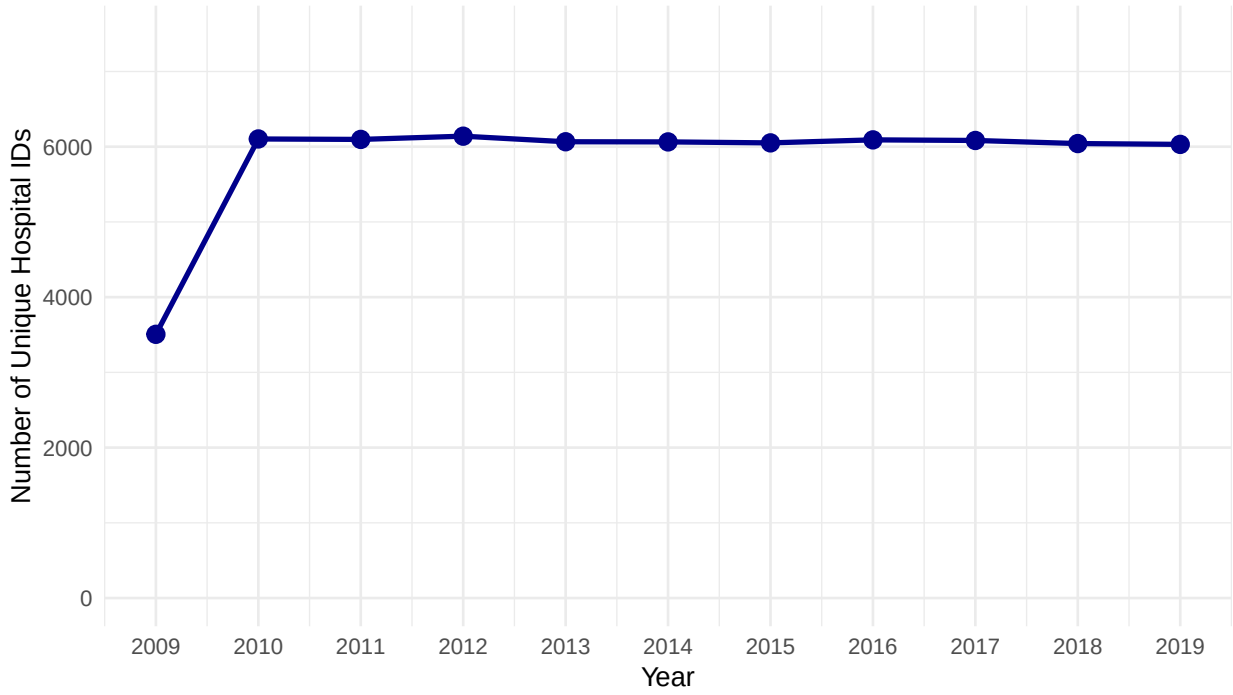


Figure 2: Number of Unique Hospital IDs Over Time (2009-2019)

After cleaning the raw data files for versions 1996 and 2010 to include only unique reports, I counted the number of unique provider numbers, which corresponds to unique hospitals. In total, there are **6,861** unique hospital IDs in the data. Figure 2 presents changes in the number of unique hospital IDs over time. While there is a dramatic increase from 2009 to 2010, the number of unique IDs appears to remain stable afterwards.

Question 3

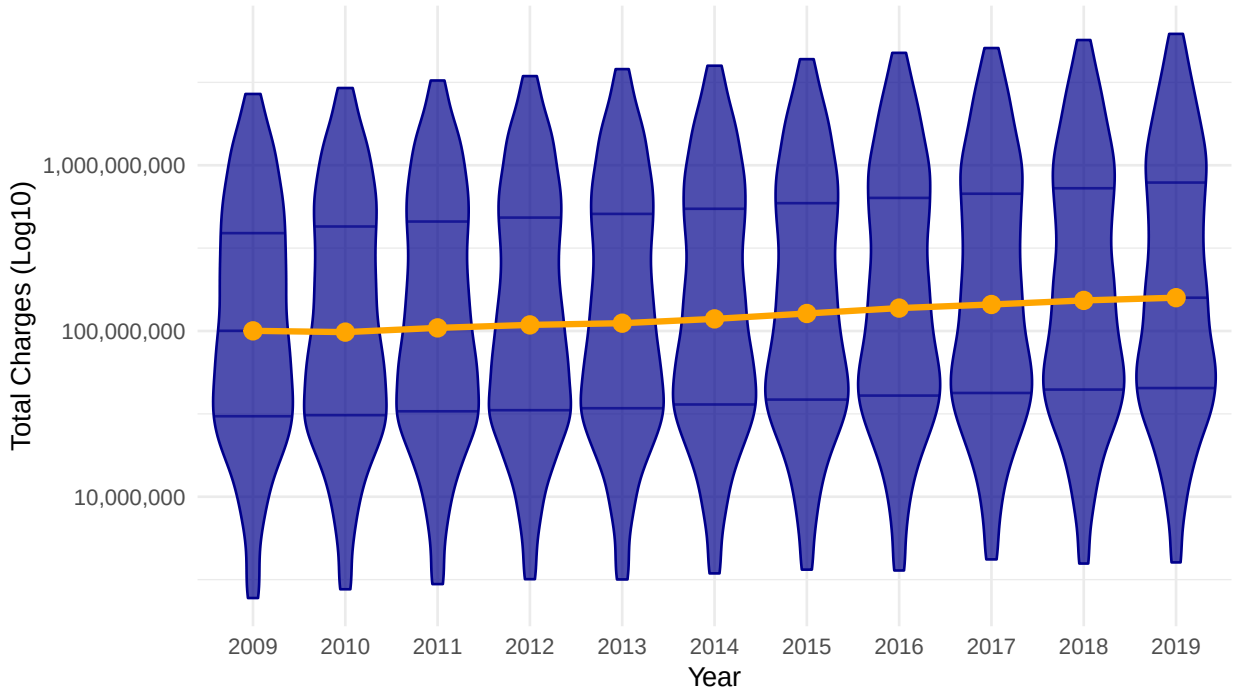


Figure 3: Distribution of Total Charges Per Year (2009-2019)

Figure 3 presents the distribution of total charges per year, excluding the top and bottom 1%. The median of distributions per year are marked in orange, showing a very small increase in total charges over time.

Question 4

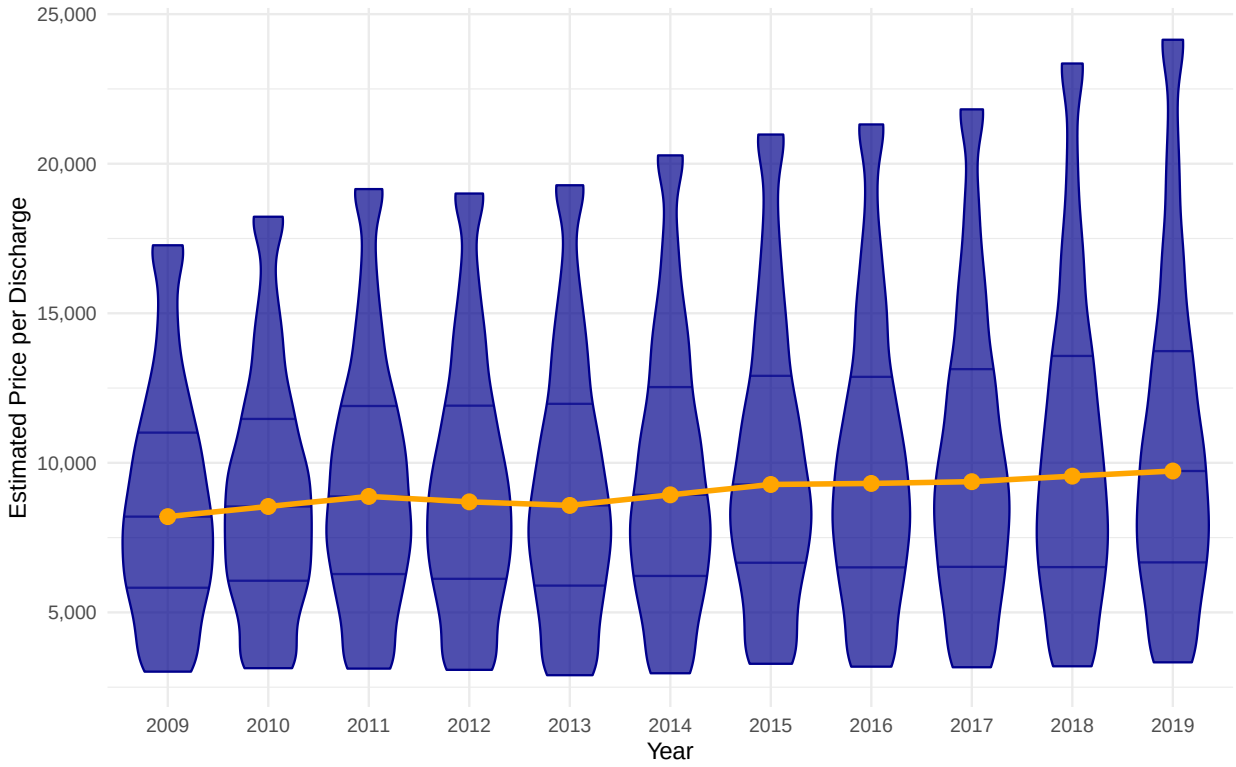


Figure 4: Distribution of Estimated Hospital Prices By Year (2009-2019)

Figure 4 presents the distribution of estimated prices per year, excluding the top and bottom 5%. Impossible values, such as hospitals having more Medicare discharges than total discharges, were also excluded. While the median estimated prices appear to be stable over time, the maximum prices are observed to increase.

Question 5

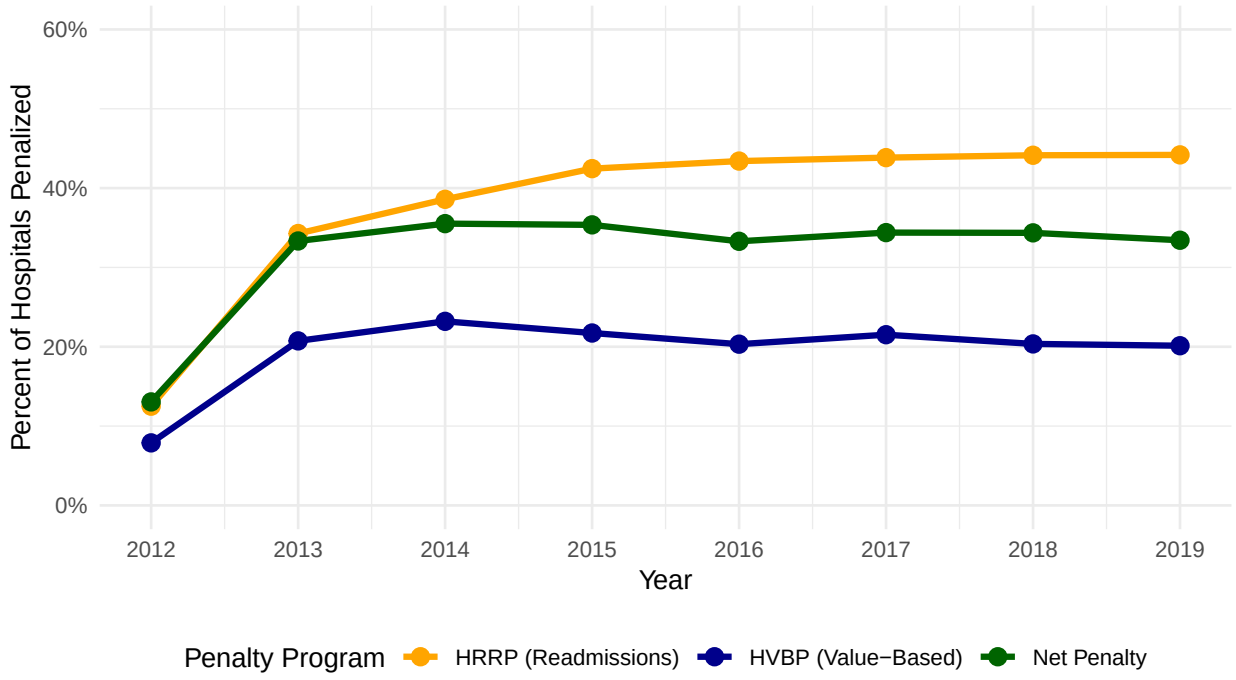


Figure 5: Share of Hospitals Penalized (2012-2019)

Figure 5 presents the share of hospitals that were penalized from 2012 to 2019. This is stratified such that the percentage of hospitals affected by HRRP, VBP, or both are shown.

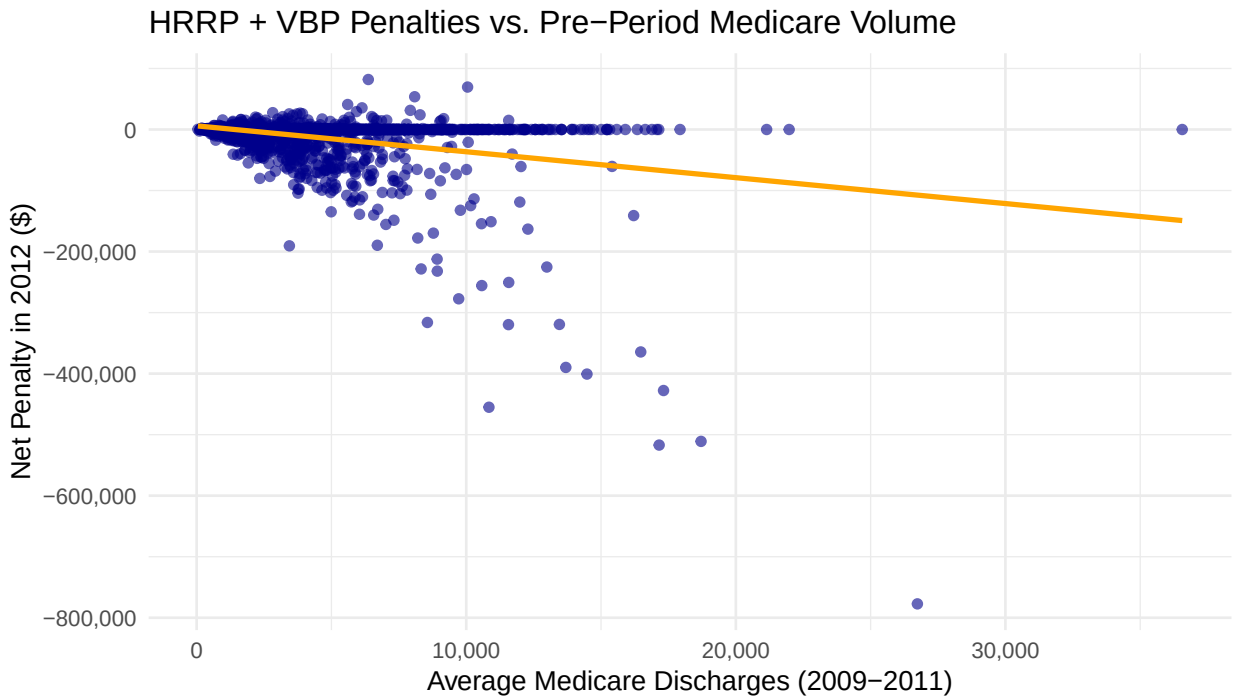
Question 6

Table 1: OLS Estimates of Net Penalty on Price Changes

	Baseline	+ Beds	+ Medicaid
Intercept	209.314***	191.613***	36.056
	(54.317)	(54.887)	(67.402)
Net Penalty	−0.003*	−0.003*	−0.002+
	(0.001)	(0.001)	(0.001)
Avg. Beds		0.074*	0.064+
		(0.034)	(0.034)
Avg. Medicaid Discharges			0.101***
			(0.025)
Num.Obs.	2435	2435	2425
R2	0.002	0.004	0.011

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Question 7



Question 8

Table 2: First Stage: 2012 Net Penalty on Pre-Penalty Medicare Volume

	Baseline	+ Beds	+ Medicaid
Intercept	6011.603***	6009.156***	5464.820***
	(1140.485)	(1141.175)	(1156.964)
Avg. Medicare Discharges	−4.245***	−4.247***	−4.699***
	(0.239)	(0.240)	(0.277)
Avg. Beds		0.037	−0.005
		(0.483)	(0.483)
Avg. Medicaid Discharges			1.346**
			(0.411)
Num.Obs.	2435	2435	2425
R2	0.115	0.115	0.119

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 3: Reduced Form: Price Change (2011-2014) on Pre-Penalty Medicare Volume

	Baseline	+ Beds	+ Medicaid
Intercept	172.088*	167.199*	115.837
	(80.720)	(80.693)	(81.563)
Avg. Medicare Discharges	0.019	0.015	−0.027
	(0.017)	(0.017)	(0.020)
Avg. Beds		0.073*	0.069*
		(0.034)	(0.034)
Avg. Medicaid Discharges			0.127***
			(0.029)
Num.Obs.	2435	2435	2425
R2	0.000	0.002	0.010

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Question 9

Table 4: 2SLS Estimates: Effect of Net Penalty on Price Changes (2011-2014)

	Baseline	+ Beds	+ Medicaid
Intercept	198.414**	188.159**	83.875
	(64.516)	(64.647)	(71.621)
Net Penalty	−0.004	−0.003	0.006
	(0.004)	(0.004)	(0.004)
Avg. Beds		0.073*	0.069*
		(0.034)	(0.034)
Avg. Medicaid Discharges			0.119***
			(0.027)
Num.Obs.	2435	2435	2425

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Question 10

When using prior Medicare discharges as the study instrument, the "local" ATE (LATE) measures the effect for "compliers," or hospitals whose penalty was determined by their pre-2012 Medicare volume, on the change in price per discharge (2011-2014). On the other hand, the ATE obtained from the OLS regression models measure the effect of the treatment, 2012 net penalties, on the aforementioned outcome. There are potentially two main drivers for the difference in estimates. First, the OLS regression understates the true effect by including wealthy, specialty hospitals that

can easily absorb penalties without changing prices. Meanwhile, the LATE specifically capturing hospitals with large Medicare volumes implies larger estimates, since these hospitals do not have large cash reserves and must raise prices to continue operations. Second, the treatment is endogenous: penalties affect prices, and the opposite may be true when considering that large investments in quality improvement can reduce readmission rates.